Autonomous Systems and Robotics 700 Assignment

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Student ITS No: _____

Qualification: _____ Year of Study: _____ Semester: _____

Assignment due date: _____ Date submitted: _____

QUESTION	EXAMINER MARKS	MODERATOR MARKS	REMARKS

ASSIGNMENT INSTRUCTIONS

Please tick each box to confirm completion.

Use Times New Roman font, size 12, with 1.5 line spacing throughout the document. Apply Harvard Referencing Style for all citations and references.

For essay-style assignments, please include the following sections:

Table of Contents

Introduction

Main Body (with relevant subheadings)

Conclusion

References

Submit the assignment in PDF format on Moodle.

Use the specified cover page provided.

Include a signed declaration of originality.

DECLARATION OF ORIGINALITY:

I hereby declare that this assignment is my own work and has not been copied from any other source except where due acknowledgment is made. I affirm that all sources used have been properly cited and that this submission complies with the institution's policies on academic integrity and plagiarism.

Student Signature: **Date:**

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Table of Contents

INTRODUCTION 2
METHODOLOGY 2
COMPONENT A: 2
COMPONENT B: 3
RESULTS & EVALUATION 4
DISCUSSION & ETHICAL CONSIDERATIONS 7
CONCLUSION & RECOMMENDATIONS 8

INTRODUCTION

Autonomous robotic systems integrate perception, planning and feedback control to function in environments that may be unpredictable or shared with people. This project addresses the assignment's two complementary tasks: A- is a single-robot perception, motion planning and control while B is about cooperative multi-agent search and rescue. The assignment does not use ROS or Gazebo, but it does require Webots for modelling and simulation, Matplotlib for visualizations.

The local navigation problem is represented by Component A, which involves controlling a differentia-drive rover, interpreting sensor observations and planning from a start posture to a goal while avoiding obstacles. Component B represents avocation and coordination which distributes targets across three agents, plan collision-free routes and integrates local progress estimates. When taken as a whole, the system show that a shortest path alone is insufficient because human safety, communication breakdowns, controller responsiveness, and perception errors all affect system performance.

METHODOLOGY

COMPONENT A:

1. ENVIRONMENT AND SENSOR DESIGN

The given indoor_navigation.wbt world is tagged Webots R2025a version and uses a north-up-east coordinate system. It defines a 12m × 12m, dark permitter walls, nine static concrete barriers, a rover start at approximately (-5.16, -5.16) m and a goal marker at (4.65, 1.50) m. The single robot node has a camera with recognition enabled (160 × 100 pixels; field of view 1.431 rad), a 180-sample single-layer LiDAR with a 360°-horizontal field of view and 4.2m maximum range, GPS, compass, two-wheel motors and position sensors.

An appropriate perception pipeline would convert LiDAR return into local obstruction points, would use calibrated camera geometry to project camera detections into the ground plane and then combine both sources into a single occupancy grid. LiDAR is helpful for geometric barrier boundaries and provides metric range; camera recognition can identify semantic aspects that range alone cannot. Filtering invalid/ near range readings and time-aligning sensor samples would reduce spurious cells. These design suggestions are based on the configured devices; their implementation and calibration cannot be verified because the controller source is absent.

2. ROVER KINEMATICS

The wheel radius in the world is 0.05m and the wheel anchors are separated by approximately 0.30m. A differential-drive model is therefore $v = r(\omega R + \omega L)/2$ and $\omega = r(\omega R - \omega L)/b$, where r is wheel radius and b is track width. The robot pose evolves as $\dot{x} = v \cos\theta$, $\dot{y} = v \sin\theta$, $\dot{\theta} = \omega$. The world sets a maximum motor velocity of 6.28 rad/s; no tuning results are presented because no PID gain sor response date were provided.

3. PLANNING and CLOSED-LOOP CONTROL

The assignment calls for Dijkstra and A* on the same occupancy grid, comparing cost, nodes explored and computation time, plus closed-loop PID tracking. Dijkstra is optimal with non-negative edge weights; A* adds a heuristic and remains optimal with a consistent heuristic. Grid resolution and start/goal conditions must remain constant for a meaningful comparison. The completeness of this comparison cannot be confirmed since the Component A controller and run outputs were not provided.

COMPONENT B:

4. SWARM SETUP and TASK ALLOCATION

The Component B world defines three coloured Robot nodes which is Robot 1 the leader or explorer, Robot 2 the explorer and Robot 3 the searcher, and the three targets are in an 18m × 14m, courtyard with four obstacles. One supervisor controller is in charge of its 64ms Webots step. Instead of operating simulated wheel motors, the controller directly modifies the robots' positions; this is a repeatable kinematic abstraction rather than a physical differential drive or sensor simulation.

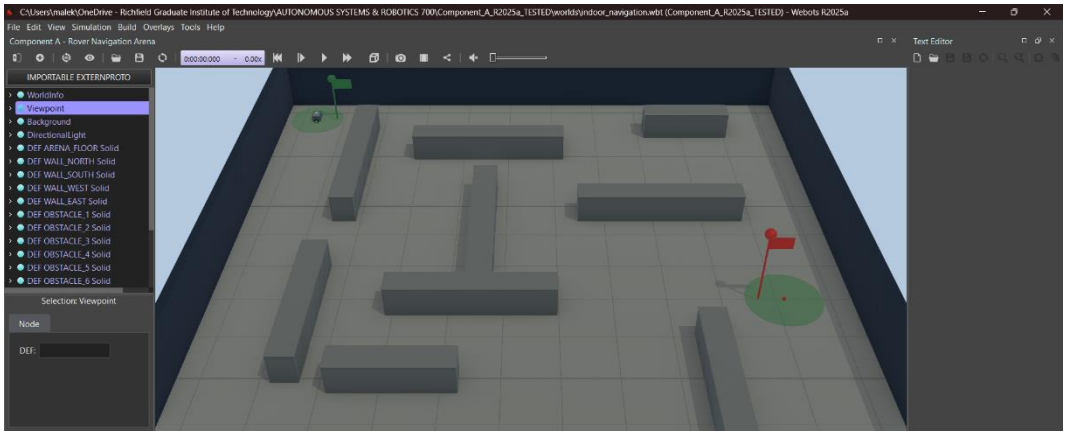
PSO is a population-based stochastic optimizer introduced by Kennedy and Eberhart (1995); in this small deterministic geometry, its objective is transparent but does not include obstacle aware route length until after the assignment. Task allocation uses random-key discrete PSO. The goal is the total of the Euclidean start-to-target distances; each particle encodes a permutation of victim assignments. The implementation uses 32 particles and 55 iterations. The observed best assignment is Robot 1 -> Victim 1, Robot 2-> Victim 2 and Robot -> Victim 3.

5. CONSENSUS and LEARNING POLICY

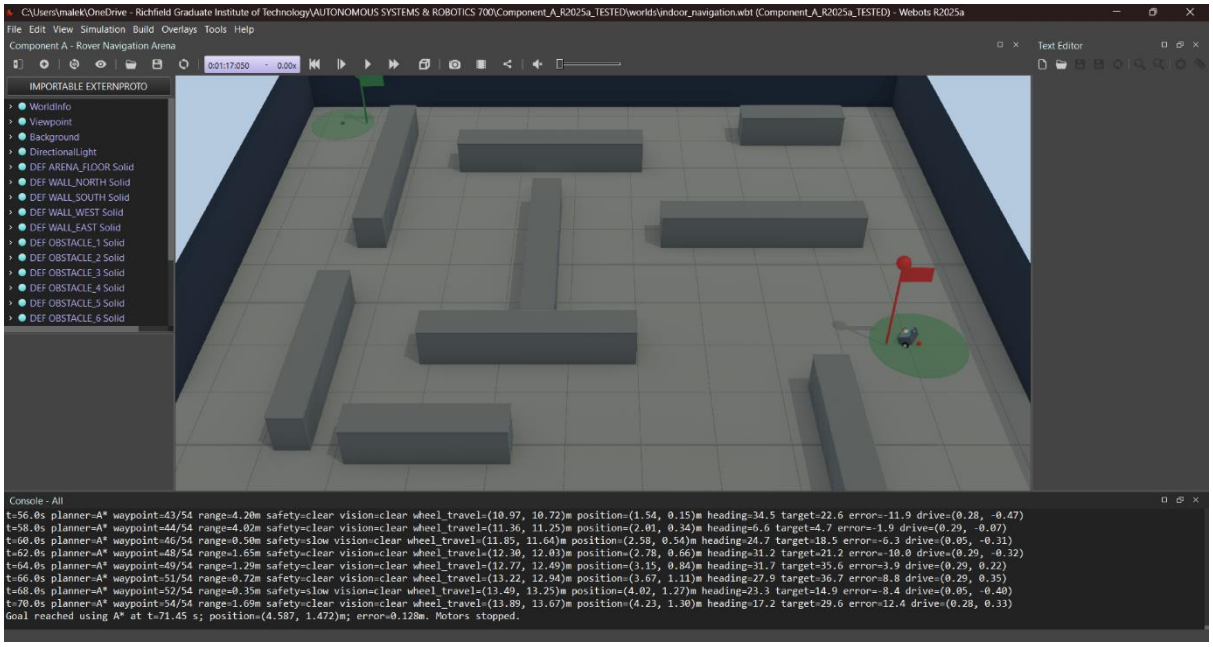
Robot 1 is connected to both searchers by logical star. Each synchronous update blends neighbour estimates with local normalized progress; this is not a radio or packet-loss model. The controller trains a 3-6-2 neural network on 450 generated samples for 100 epochs. Goal distances, obstacle clearance and route progress are the inputs; labels adhere to the guideline "cautious if distance < 1.m or clearance < 0.65m." For each simulation phase, outputs choose a smaller or larger pose increment. Instead of gaining rescue knowledge, from field data, the network approximates its synthetic rule.

RESULTS & EVALUATION

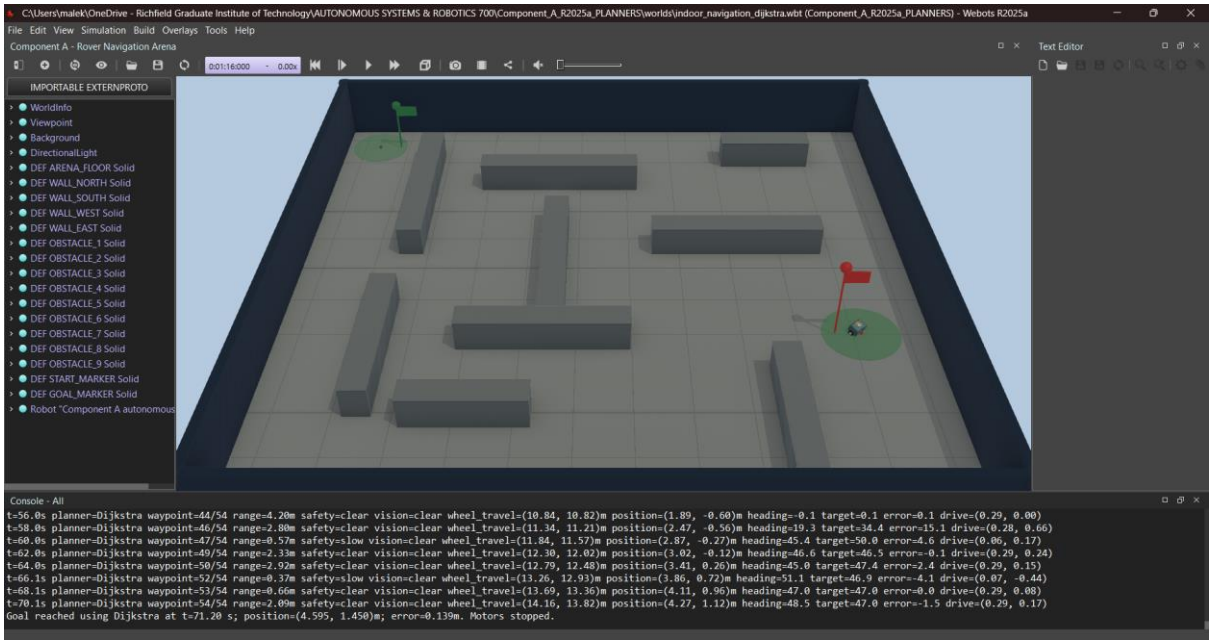
COMPONENT A:



The robot in this figure is at stationary which means it is not moving. The green flag represents the start point and the red flag represent the end or the goal. The robot has to figure its way from green to red flag while avoiding obstacles.

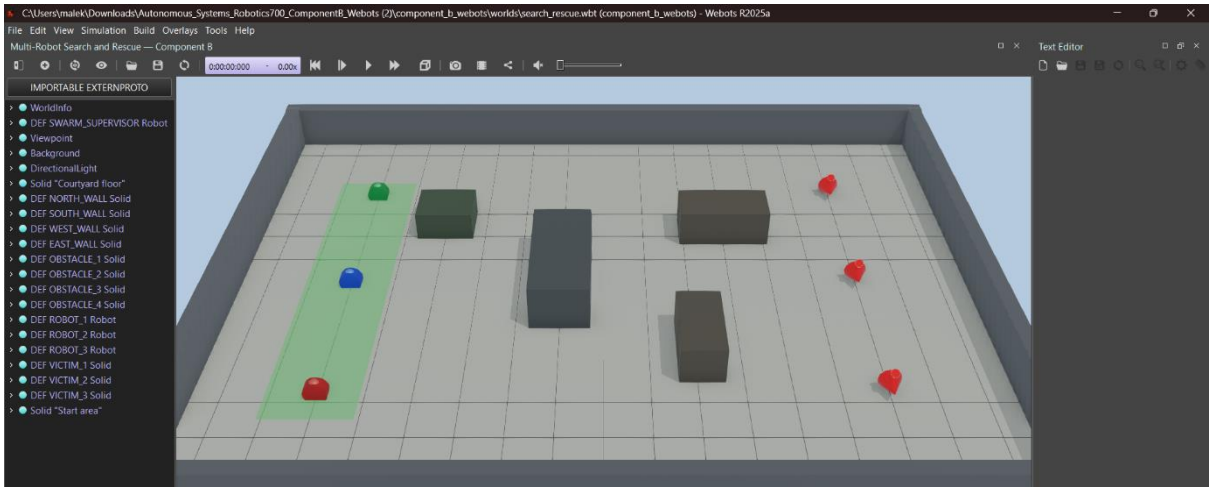


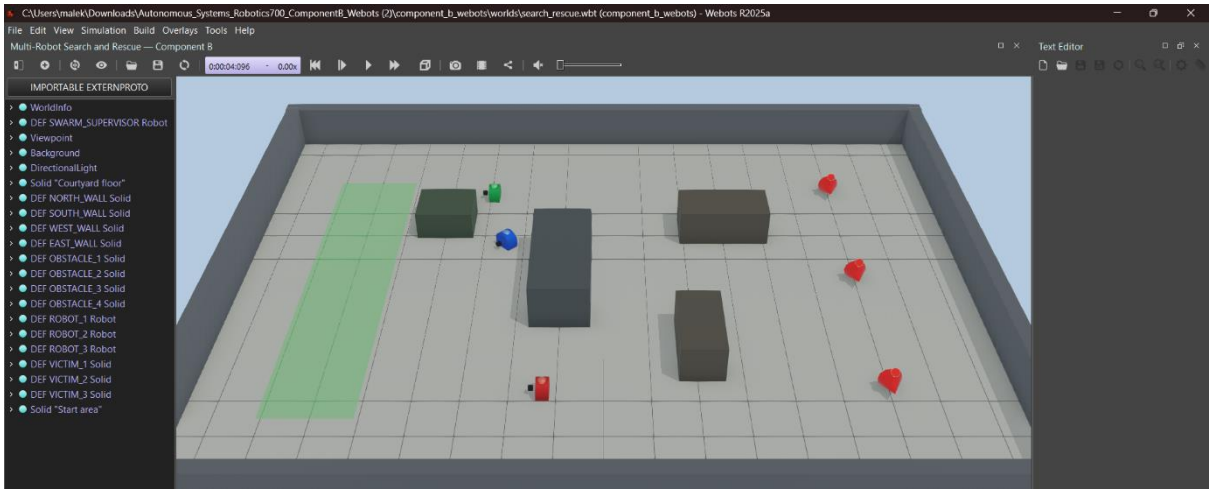
The robot has figured its way from green to red by using A* Algorithm.

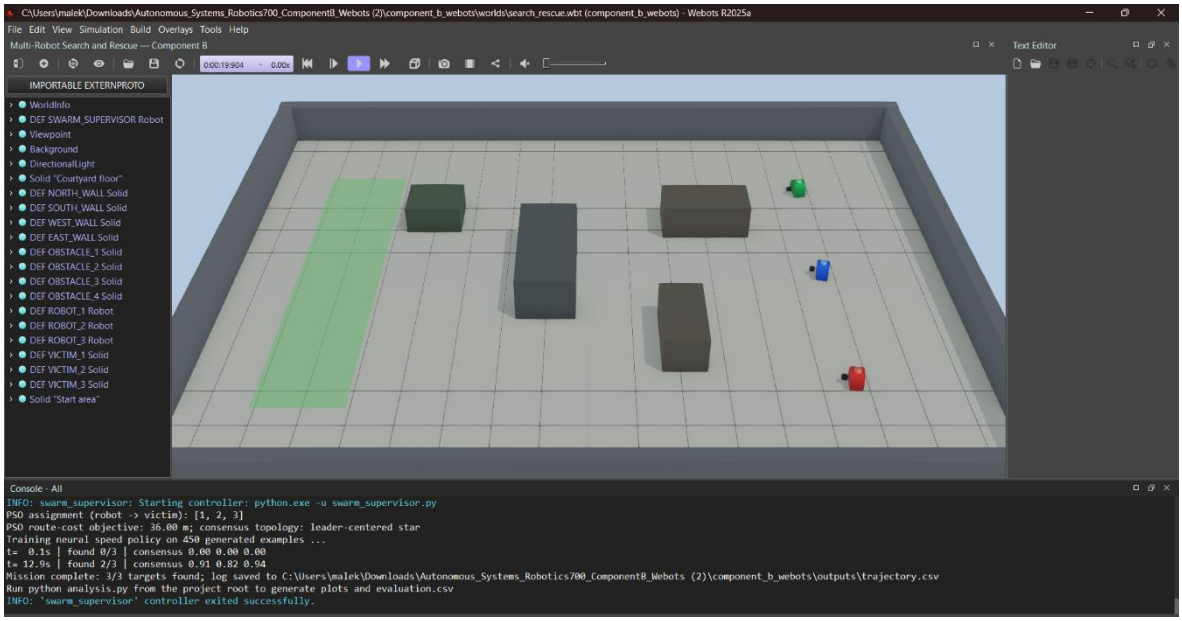


The robot has reached its goal by using Dijkstra Algorithm.

COMPONENT B:







The three figures show the robots at stationary, when they start moving and choosing their path until they reach their goals. The three robots communicate with one another to create a swift environment and do not crash into one another.

PERFORMANCE SUMMARY

MEASURE	**OBSERVED or COMPUTED VALUE**	**INTERPRETATION/ LIMITATION**
Component A sensor/ world setup	12m × 12m; camera, LiDAR, GPS, compass; 9 barriers	Verified from WBT configuration; controller behaviour unverified.
Component A PID/ planner metric	Not available	No controller source, trajectory or response logs supplied
PSO assignment objective	36.00m	Console value; straigjt-line assignment cost, not path length
Planned A* distance (Component B)	37, 68m total	Offline recomputation from supplied planner and map; not a measured run trace
Targets found (Component B)	3/3	Found in user-supplied Webots console screenshot
Neural policy hold-out accuracy	98.4% (n = 1,000)	Offline synthetic labels; same threshold rule as training, not field or Webots-run validation

DISCUSSION & ETHICAL CONSIDERATIONS

Integration, trade-offs and decision architecture

Component A's configured LiDAR-camera iteration may enhance obstacle representation, but only if coordinate calibration, timing and detection uncertainty are managed. A cautious occupancy grid can lengthen routes, close off tight spaces and lower the danger of collisions. A* can reduce search effort relative to Dijkstra when the heuristic is informative; either planner's route remains dependent on map quality. While conservative gains lessen overshot at the expense of delayed response, responsive controllers may follow corners rapidly but increase measurement noise. These trade-offs require measured curves; the supplied component A evidence is insufficient to rank them.

Component B currently makes centralized decisions inside a single Supervisor. This provides a global perspective and makes PSO assignment and simplifies deterministic replay, but the Supervisor is a single point of failure and does not demonstrate genuinely distributed execution. Each robot in a distributed design would keep a local planner and only communicate about target claims, progress and hazards. Stale state, conflicting assignments, message loss and consensus delay are introduced, while graceful degradation and scalability are improved. The current star graph is not a fault-tolerant communication system, but rather a reasonable teaching baseline.

Risk Controls

COMPONENT A:

FAILURE MODE	**POTENTIAL CONSEQUENCE**	**MITIGATION BEFORE DEPLOYMENT**
Obstacle missed	Collision/ unsafe route	Calibrate sensors; inflate obstacles; independent stop-distance
Supervisor / link failure	Stop, drift or duplicate search	Watchdog, safe stop, local fallback, heartbeat and reassignment timeout.
NN misclassification	Excess speed near hazards	Hard speed bounds, external safety envelope, uncertainty monitor and manual override.
Personal data exposure	Privacy harm/ reuse	Minimize retention, restrict access, state purpose and deletion rules, review privacy.

COMPONENT B: has no sensor noise, motor dynamics, wheel slip, packet loss or people. The NN uses synthetic labels and behaviour outside that distribution is unknown. Retain logged inputs/ actions, rule-based reason codes and a deterministic safety override for auditability.

CONCLUSION & RECOMMENDATIONS

Component A and Component B are established with different designs. Component A is a sensor-equipped differential-drive rover and comprehensive indoor arena but its navigation controller and performance data are missing from the materials reviewed. Component B is a reproducible three-agent kinematic simulation with PSO task assignment, A* route generation, a leader-centred consensus update and a rule-trained neural speed policy. According to the provided Webots console screenshot, all three targets were located. The evidence that is currently available show to a successful demonstration in the simplified simulated world rather than deployment readiness.